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Real Mixed Hodge Structures

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Real mixed Hodge structures (§2) form a rigid abelian $\otimes$-category, and the functor "underlying real vector space" is a fiber functor. The functor Gr^W is another. We propose to compute what the general theory of Tannakian categories gives in this particular case.

1 Three opposed filtrations

1.1

This section is a variation on Section 1.2 of P. Deligne, *Théorème de Hodge II*, Publ. Math. IHES **40**, pp. 5–58. Let $\mathcal{A}$ be an abelian category, and let A be an object of $\mathcal{A}$ equipped with three opposed filtrations W, F, $\bar{F}$. Here "filtration" means "finite filtration" (loc. cit. 1.1.4); the filtration W is assumed increasing, and the filtrations F, $\bar{F}$ decreasing. The opposition condition is written

$$\mathrm{Gr}_F^p \mathrm{Gr}_{\bar{F}}^q \mathrm{Gr}_n^W A = 0 \quad \text{for } n \neq p+q$$

(loc. cit. 1.2.7, 1.2.13).

For every bigraded object $B = \bigoplus B^{p,q}$, we shall write W, F_W, and $\bar{F}_W$ for the three opposed filtrations

$$W_n = \bigoplus_{p+q \leq n} B^{p,q},$$
$$F_W^i = \bigoplus_{p \geq i} B^{p,q},$$
$$\bar{F}_W^i = \bigoplus_{q \geq i} B^{p,q}.$$

With this notation, to say that the three filtrations W, F, and $\bar{F}$ of A are opposed is equivalent (loc. cit. 1.2.6) to saying that each $\mathrm{Gr}_n^W A$ admits a unique decomposition

$$\mathrm{Gr}_n^W A = \bigoplus_{p+q=n} A_W^{p,q}$$

such that the filtrations of $\mathrm{Gr}_n^W A$ induced by F and $\bar{F}$ are given by

$$F^i \mathrm{Gr}_n^W A = \bigoplus_{\substack{p+q=n \\ p \geq i}} A_W^{p,q},$$
$$\bar{F}^i \mathrm{Gr}_n^W A = \bigoplus_{\substack{p+q=n \\ q \geq i}} A_W^{p,q}.$$

In other words, F and $\bar{F}$ induce on $\mathrm{Gr}_n^W A$ the filtrations F_W and $\bar{F}_W$.

Set

$$A_F^{p,q} := (W_{p+q} \cap F^p) \cap \Big((W_{p+q} \cap \bar{F}^q) + \sum_{i>0} (W_{p+q-i} \cap \bar{F}^{q+1-i}) \Big)$$

and, interchanging the roles of F and $\bar{F}$ and of p and q,

$$A_{\bar{F}}^{p,q} := (W_{p+q} \cap \bar{F}^q) \cap \Big((W_{p+q} \cap F^p) + \sum_{i>0} (W_{p+q-i} \cap F^{p+1-i}) \Big).$$

For $p + q = n$, the projection of W_n onto $\mathrm{Gr}_n^W A$ induces an isomorphism from $A_F^{p,q}$ to $A_W^{p,q}$: loc. cit. 1.2.8, restricted to the special case 1.2.11, where $A_F^{p,q}$ is denoted $A_0^{p,q}$; note that in the formula defining $A_F^{p,q}$, one may replace the sum over $i > 0$ by the same sum over $i \geq 2$. It follows (loc. cit.) that the $A_F^{p,q}$, respectively the $A_{\bar{F}}^{p,q}$, form a bigrading of A, and that the sum a_F, respectively $a_{\bar{F}}$, of the isomorphisms

$$A_F^{p,q} \xrightarrow{\sim} A_W^{p,q}, \qquad A_{\bar{F}}^{p,q} \xrightarrow{\sim} A_W^{p,q}$$

is a bifiltered isomorphism from $(A; W, F)$ to $(\mathrm{Gr}^W A; W, F_W)$, respectively from $(A; W, \bar{F})$ to $(\mathrm{Gr}^W A; W, \bar{F}_W)$.

Lemma 1.1. *The automorphism $d = a_{\bar{F}} a_F^{-1}$ of $\mathrm{Gr}^W A$ satisfies*

$$(d-1)(A_W^{p,q}) \subset \bigoplus_{\substack{r<p \\ s<q}} A_W^{r,s}. \tag{1.1.1}$$

Passing to the associated graded, the isomorphisms a_F and $a_{\bar{F}}$ induce the identity of $\mathrm{Gr}^W A$. Hence $\mathrm{Gr}^W(d) = 1$.

For each subset I of $\mathbb{Z}^2$, and for B a bigraded object, set $B_I = \bigoplus_{(p,q) \in I} B^{p,q}$. Set

$$\begin{aligned} I(p,q) &= \{(r,s) \mid r+s \leq p+q \text{ and } ((r,s) = (p,q) \text{ or } r < p)\}, \\ J(p,q) &= \{(r,s) \mid r+s \leq p+q \text{ and } ((r,s) = (p,q) \text{ or } s < q)\}, \\ K(p,q) &= \{(r,s) \mid r+s \leq p+q \text{ and } r \geq p\}. \end{aligned}$$

$$\begin{array}{lcc}
 & \begin{array}{ccccl} & & & \times & \\ & & \times & \times & \\ & \times & \times & \circ & (p,q) \\ \circ & & & \circ & \\ \circ & \circ & & \circ & \end{array} & \begin{array}{ccccl} & & & \circ & \\ & & \circ & \circ & \\ & \circ & \circ & \times & (p,q) \\ \times & & & \circ & \\ \times & \times & & \circ & \end{array} \\
I(p,q): & & \\
J(p,q): & & \\
K(p,q): \text{ circled} & &
\end{array}$$

In the definition of $A_F^{p,q}$ as an intersection of two subobjects of A, the first subobject is $A_F^{I(p,q)}$ and the second is $A_{\bar{F}}^{J(p,q)}$. This follows from the fact that a_F, respectively $a_{\bar{F}}$, is a bifiltered isomorphism. Thus $A_F^{p,q} \subset A_F^{I(p,q)}$, whence

$$dA_F^{p,q} = a_{\bar{F}} a_F^{-1} A_F^{p,q} \subset A_{\bar{F}}^{I(p,q)},$$

and $dA_F^{p,q} \subset A_{\bar{F}}^{J(p,q)}$. Since $\mathrm{Gr}^W(d) = 1$, one has equality $dA_F^{I(p,q)} = A_{\bar{F}}^{I(p,q)}$. Interchanging F and $\bar{F}$, one similarly obtains $d^{-1} A_{\bar{F}}^{J(p,q)} = A_F^{J(p,q)}$, hence $dA_F^{J(p,q)} = A_{\bar{F}}^{J(p,q)}$, and

$$dA_W^{p,q} = dA_W^{I(p,q)} \cap dA_W^{J(p,q)} = A_W^{I(p,q) \cap J(p,q)} = A_W^{K(p,q)} = \bigoplus_{\substack{r<p \\ s<q}} A_W^{r,s}.$$

We conclude by using $\mathrm{Gr}^W(d) = 1$.

Proposition 1.2. *The functor $A \mapsto \operatorname{Gr}^W A$ is an equivalence from the category of objects of $\mathcal{A}$ equipped with three opposed filtrations W, F, $\bar{F}$, to the category of objects of $\mathcal{A}$ equipped with a finite bigrading and an automorphism d satisfying* (1.1.1)*.*

The isomorphism $a_F : A \xrightarrow{\sim} \operatorname{Gr}^W A$ respects the filtration W, carries F to F_W, and carries $\bar{F}$ to $d^{-1}(F_W)$: one has $a_{\bar{F}} = d^{-1} a_F$. Thus the functor Gr^W has a left inverse, namely the functor

$$s : (B = \bigoplus B^{p,q}, d) \longmapsto (B, W, F_W, d^{-1}(F_W)),$$

equipped with the functorial isomorphism $a_F : A \xrightarrow{\sim} s \operatorname{Gr}^W A$.

Conversely, start from a bigraded object B equipped with an automorphism d, and define W, F, and $\bar{F}$ as above. These filtrations are opposed as soon as $\operatorname{Gr}^W(d) = 1$. For these filtrations,

$$B_F^{p,q} = B^{K(p,q)} \cap d^{-1}(B^{I(p,q)}).$$

If d satisfies (1.1.1), then $dB^{p,q} \subset B^{I(p,q)}$ and $d^{-1}B^{p,q} \subset B^{J(p,q)}$. Since $(B^{p,q})$ and $(B_F^{p,q})$ are bigradings of B, one has equality $B_F^{p,q} = B^{p,q}$. The same argument, applied to d^{-1} and to the bigrading $\bar{F}^{p,q}$, shows that $d^{-1}B^{\bar{F}^{p,q}} = B^{p,q}$. It follows that the evident isomorphism from B to $\operatorname{Gr}^W B$ is an isomorphism of functors $\mathrm{Id} \xrightarrow{\sim} \operatorname{Gr}^W \circ s$.

Remark 1.3. If 2 is invertible, d has a unique square root $d^{1/2}$ which still satisfies (1.1.1). Set

$$a := d^{1/2} a_F = d^{-1/2} a_{\bar{F}}.$$

Then $a_{\bar{F}} = d^{1/2} a$ and $a_F = d^{-1/2} a$, so that a carries F, respectively $\bar{F}$, to the filtration $d^{1/2}(F_W)$, respectively $d^{-1/2}(\bar{F}_W)$, of $\operatorname{Gr}^W A$. The isomorphism a is the same for A equipped with $W, F, \bar{F}$ and for A equipped with $W, \bar{F}, F$.

1.2

These constructions are compatible with tensor product. If $\otimes : \mathcal{A}_1 \times \mathcal{A}_2 \to \mathcal{A}$ is an exact biadditive functor, we extend it to filtered objects by setting

$$F^n(A_1 \otimes A_2) = \sum_{a+b=n} F^a(A_1) \otimes F^b(A_2) \subset A_1 \otimes A_2.$$

The associated graded functor is compatible with $\otimes$, and for bifiltered objects the isomorphism

$$\operatorname{Gr}^{W_1 \otimes W_2}(A_1 \otimes A_2) \xrightarrow{\sim} \operatorname{Gr}^{W_1}(A_1) \otimes \operatorname{Gr}^{W_2}(A_2)$$

carries the filtration induced by $F_1 \otimes F_2$ to the tensor product of the filtrations induced by F_1 and F_2.

For vector spaces over a field, this can be deduced from the fact that a bifiltration always underlies a bigrading. It follows that if W_i, F_i, and $\bar{F}_i$ are three opposed filtrations on A_i for $i = 1, 2$, then the filtrations $W_1 \otimes W_2$, $F_1 \otimes F_2$, and $\bar{F}_1 \otimes \bar{F}_2$ on $A_1 \otimes A_2$ are again opposed.

The formation of the bigradings $A_F^{p,q}$ and $A_{\bar{F}}^{p,q}$ of A, and $A_W^{p,q}$ of $\operatorname{Gr}^W(A)$, as well as the isomorphisms $a_F, a_{\bar{F}} : A \xrightarrow{\sim} \operatorname{Gr}^W(A)$ and the automorphism d of $\operatorname{Gr}^W(A)$, is compatible with $\otimes$.

1.3

Let k be a commutative field, let $\mathcal{A}$ be the category of finite-dimensional vector spaces over k, and let $\mathscr{C}$ be the category of vector spaces equipped with three opposed filtrations W, F, and $\bar{F}$. This is a rigid abelian $\otimes$-category with $\mathrm{End}(1) = k$. It has the fiber functor ω_0, the underlying vector space functor, and hence is neutral Tannakian. We also have the fiber functor $\mathrm{Gr}^W =: \omega_W$. The isomorphisms a_F and $a_{\bar{F}}$ are isomorphisms of fiber functors $\omega_0 \xrightarrow{\sim} \omega_W$. Put $G = \mathrm{Aut}(\omega_W)$. The functorial automorphism d defines a k-point of the pro-algebraic group G.

The functorial bigrading of $\mathrm{Gr}^W(V)$ by the $V_W^{p,q}$ corresponds to a homomorphism

$$\alpha : \mathbb{G}_\mathrm{m} \times \mathbb{G}_\mathrm{m} \longrightarrow G.$$

It is a section of the projection

$$\mathrm{pr} : G \to \mathbb{G}_\mathrm{m} \times \mathbb{G}_\mathrm{m}$$

corresponding to the inclusion in $\mathscr{C}$ of the category of bigraded vector spaces, via

$$B \longmapsto (B, W, F_W, \bar{F}_W).$$

I wish to normalize α so that $\alpha(\lambda, \mu)v = \lambda^p \mu^{-q} v$ for $v \in V_W^{p,q}$.

By Proposition 1.2, G is also the algebraic group of automorphisms of the underlying-vector-space fiber functor on the category of bigraded vector spaces equipped with an automorphism d satisfying (1.1.1). We are going to "compute" it when k has characteristic 0.

Let $\mathcal{L}$ be the Lie algebra over k freely generated by elements $D^{i,j}$ for $i, j < 0$, and equip it with the unique bigrading for which $D^{i,j}$ has bidegree (i, j). The $W_n(\mathcal{L})$ are ideals of $\mathcal{L}$, and the quotients $\mathcal{L}/W_n(\mathcal{L})$ are nilpotent Lie algebras. We shall write U for the pro-algebraic group which is the projective limit of the unipotent algebraic groups $\exp(\mathcal{L}/W_n(\mathcal{L}))$ with Lie algebras $\mathcal{L}/W_n(\mathcal{L})$. Let $\mathbb{G}_\mathrm{m} \times \mathbb{G}_\mathrm{m}$ act on $\mathcal{L}$ so that (λ, μ) acts on an element D of degree (i, j) by multiplication by $\lambda^i \mu^j$. This action respects the filtration W and therefore gives an action of $\mathbb{G}_\mathrm{m} \times \mathbb{G}_\mathrm{m}$ on U.

Construction 1.4. The group G is the semidirect product $(\mathbb{G}_\mathrm{m} \times \mathbb{G}_\mathrm{m}) \ltimes U$.

To make the semidirect product act on a finite-dimensional vector space V is the same as to make $\mathbb{G}_\mathrm{m} \times \mathbb{G}_\mathrm{m}$ and U act, subject to a compatibility condition relating the two actions. The action of $\mathbb{G}_\mathrm{m} \times \mathbb{G}_\mathrm{m}$ corresponds to a bigrading, namely the one for which $(\lambda, \mu)v^{p,q} = \lambda^p \mu^{-q} v^{p,q}$. The action of U corresponds to an action of $\mathcal{L}$, zero on some W_n and nilpotent, i.e. to the data of endomorphisms $D^{i,j}$, for $i, j < 0$, such that every composite of $D^{i,j}$ of sufficiently large absolute weight is zero. Compatibility says that $D^{i,j}$ has bidegree (i, j), and this property implies the required vanishing just stated. The data of the $D^{i,j}$ is equivalent to the data of $D := \sum D^{i,j}$. Altogether, the action of the semidirect product on V is identified with the data of a bigrading and of an endomorphism D of V satisfying

$$DV^{p,q} \subset \bigoplus_{i<0,\, j<0} V^{p+i,q+j}.$$

Putting $d = \exp(D)$, this is again identified with the data of a bigrading and an automorphism d satisfying (1.1.1). This construction is compatible with tensor products, giving the desired isomorphism

$$(\mathbb{G}_\mathrm{m} \times \mathbb{G}_\mathrm{m}) \ltimes U \xrightarrow{\sim} G \xrightarrow{\sim} \mathrm{Aut}(\omega_W).$$

This isomorphism is of course compatible with α and pr above.

2 Mixed Hodge structures

Recall that a real mixed Hodge structure is a finite-dimensional real vector space equipped with an increasing filtration W, whose complexification $V_{\mathbb{C}}$ is equipped with a decreasing filtration F. One requires that on $V_{\mathbb{C}}$ the filtration $W_{\mathbb{C}}$ obtained from W by extension of scalars, often denoted simply by W, together with F and its complex conjugate $\bar{F}$, form a system of three opposed filtrations.

It is equivalent to give a real vector space or to give a complex vector space equipped with a real structure, i.e. with complex conjugation, an antilinear involution. Under this dictionary, to give a real mixed Hodge structure is equivalently to give a complex vector space equipped with three opposed filtrations W, F, and $\bar{F}$, and with a complex conjugation which respects W and interchanges F and $\bar{F}$. By Proposition 1.2, the functor Gr^W induces an equivalence from the category of these objects to the category of bigraded complex vector spaces $V = \bigoplus V^{p,q}$ equipped with an automorphism d satisfying (1.1.1) and a complex conjugation c which interchanges $V^{p,q}$ and $V^{q,p}$, and for which the complex conjugate $\bar{d}$ of d is d^{-1}:

$$\bar{d} = cdc^{-1} = d^{-1}.$$

The category $\mathscr{H}$ of real mixed Hodge structures is a rigid abelian $\otimes$-category, with $\mathrm{End}(1) = \mathbb{R}$. It has the fiber functor ω_0, the underlying vector space functor, and hence is neutral Tannakian. We also have the fiber functor $\mathrm{Gr}^W =: \omega_W$. With the notation of Remark 1.3, the functorial morphism

$$a : V_{\mathbb{C}} \longrightarrow \mathrm{Gr}^W(V_{\mathbb{C}})$$

respects the real structures of both sides because it depends symmetrically on F and $\bar{F}$. It induces an isomorphism of fiber functors, again denoted

$$a : \omega_0 \xrightarrow{\sim} \omega_W.$$

With the notation of Section 1, take $k = \mathbb{C}$, and let G_2 be the real form of G defined by the following complex conjugation:

a) on $\mathbb{G}_{\mathrm{m}} \times \mathbb{G}_{\mathrm{m}}$, it is $(\lambda, \mu) \mapsto (\bar{\mu}, \bar{\lambda})$;

b) on U, it is the one compatible with the complex conjugation of $\mathcal{L}$ given by

$$D^{i,j} \mapsto -D^{j,i}.$$

As in 1.1, to make G_2 act on a real vector space V is equivalent to giving:

a) a bigrading $V_{\mathbb{C}} = \bigoplus V^{p,q}$ of $V_{\mathbb{C}}$, such that $V^{p,q}$ and $V^{q,p}$ are complex conjugates of one another, and

b) an automorphism d of $V_{\mathbb{C}}$, such that $\bar{d} = d^{-1}$ and

$$(d-1)V^{p,q} \subset \bigoplus_{\substack{i<p \\ j<q}} V^{i,j}.$$

Hence:

Proposition 2.1. *The fiber functor ω_W induces an equivalence from $\mathscr{H}$ to the category of representations of G_2: one has $G_2 = \mathrm{Aut}(\omega_W)$.*

The inverse functor Φ is obtained as follows. From a representation of G_2 on V, one obtains a bigrading of $V_{\mathbb{C}}$ and an automorphism d. To this data one attaches the real vector space V equipped with the filtration W whose complexification is the filtration W associated with the bigrading of $V_{\mathbb{C}}$, and with the filtration

$$F := d^{-1/2}(F_W)$$

of its complexification. The isomorphism $\Phi \circ \mathrm{Gr}^W \xrightarrow{\sim} \mathrm{Id}$ is the functorial isomorphism a.

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The present article reproduces, with minimal changes, the appendix bearing the same title to the preprint of E. Cattani and A. Kaplan, "On the SL(2)-orbits in Hodge theory" (IHES 1982). A different presentation of part of the results is given on pp. 470–473 of the article by E. Cattani, A. Kaplan, and W. Schmid, "Degeneration of Hodge structures" (Annals of Math. **123** (1986), 457–535).

This paper is in final form and no version of it will be submitted for publication elsewhere.